

Original Article
Editing, Writing & Publishing



Analysis of Retracted Publications on Artificial Intelligence: Trends, Ethical Concerns, and Scientific Integrity

Burhan Fatih Kocyigit ,¹ Ramazan Azim Okyay ,² Birzhan Seil ,³
Ainur B. Qumar ,⁴ and Hilmi Erdem Sumbul ,⁵

¹Department of Physical Medicine and Rehabilitation, University of Health Sciences, Adana City Research and Training Hospital, Adana, Türkiye

²Department of Public Health, Faculty of Medicine, Kahramanmaraş Sütçü İmam University, Kahramanmaraş, Türkiye

³Department of Chemical Disciplines, Biology and Biochemistry, South Kazakhstan Medical Academy, Shymkent, Kazakhstan

⁴Department of Health Policy and Management, Asfendiyarov Kazakh National Medical University, Almaty, Kazakhstan

⁵Department of Internal Medicine, University of Health Sciences, Adana Health Practice and Research Center, Adana, Türkiye



Received: Mar 11, 2025

Accepted: Apr 3, 2025

Published online: Sep 12, 2025

Address for Correspondence:

Burhan Fatih Kocyigit, MD

Department of Physical Medicine and Rehabilitation, University of Health Sciences, Adana City Research and Training Hospital, Dr. Mithat Özsan Bulvarı Kışla Mah., 4522 Sok. No: 1 Yüreğir, Adana 01230, Türkiye.
Email: bfk2701@hotmail.com

© 2025 The Korean Academy of Medical Sciences.

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

ORCID iDs

Burhan Fatih Kocyigit
<https://orcid.org/0000-0002-6065-8002>
Ramazan Azim Okyay
<https://orcid.org/0000-0001-8767-2771>
Birzhan Seil
<https://orcid.org/0000-0003-1524-8888>
Ainur B. Qumar
<https://orcid.org/0000-0003-0457-7205>
Hilmi Erdem Sumbul
<https://orcid.org/0000-0002-7192-0280>

ABSTRACT

Background: Artificial intelligence (AI) has promoted progress across various fields.

The number of papers regarding AI has risen in recent years. This study examines retracted publications regarding AI by analyzing trends, journals, and reasons.

Methods: This descriptive cross-sectional study thoroughly investigated retracted AI-related papers listed in PubMed. The data extraction comprised bibliographic data, reasons for retraction, citation metrics, journal indexing status, and Altmetric Attention Scores (AASs). Retraction notices were classified according to particular reasons. Descriptive statistics were employed to evaluate retraction trends, geographic distribution, and citation impact.

Results: A total of 764 retracted AI-related papers were examined, with the most retractions occurring in 2023 (n = 667). China had the highest number (n = 551), followed by India (n = 40) and Bangladesh (n = 23). Journals focusing on mathematical and computational biology, neurosciences, and healthcare sciences had the most retractions. The most common retraction reasons were peer review issues (n = 716) and data concerns (n = 714), followed by irrelevant citations (n = 571) and unethical AI use (n = 238). The median time to retraction was 510 days (18–4,200). The median citation and AAS scores were 0–167 and 0 (0–191).

Conclusion: The high number of retractions from China highlights the need for higher research standards. Deficits in peer review and data issues emerged as the main reasons for retraction, underscoring persistent challenges in maintaining research integrity and quality assurance. For scientific literature integrity, academic institutions, publishers, and researchers should stress transparency, ethics, and rigorous post-publication inspection.

Keywords: Artificial Intelligence; Machine Learning; Deep Learning; Retraction of Publication; Retraction Notice; Scientific Misconduct

Disclosure

The authors have no potential conflicts of interest to disclose.

Data Sharing Statement

Raw data can be provided to researchers upon request.

Author Contributions

Conceptualization: Kocyigit BF, Okyay RA, Seilil B, Kumar AB, Sumbul HE. Investigation: Kocyigit BF, Okyay RA, Seilil B, Kumar AB, Sumbul HE. Methodology: Kocyigit BF, Okyay RA, Seilil B, Kumar AB, Sumbul HE. Supervision: Kocyigit BF, Okyay RA. Writing - original draft: Kocyigit BF, Okyay RA, Seilil B, Kumar AB, Sumbul HE. Writing - review & editing: Kocyigit BF, Okyay RA, Seilil B, Kumar AB, Sumbul HE.

INTRODUCTION

Artificial intelligence (AI) has transformed multiple sectors, including healthcare, financial services, and scientific inquiry. AI has significantly accelerated innovation and decision-making through its ability to analyze extensive data, identify patterns, and generate complex outputs.^{1,2} The integration of AI technology in healthcare enhances the prediction, diagnosis, and management of disorders, ultimately benefiting patients and medical practitioners.³ The extensive utilization of AI in healthcare enhances the production of relevant studies. The corpus of pertinent literature is expanding swiftly.⁴

Medical Subject Headings (MeSH) describes AI as creating computer systems that can execute tasks typically requiring human intelligence, such as learning, problem-solving, decision-making processes, and language translation.⁵ A particular subset, Generative AI, produces synthetic material by examining data trends and utilizing deep learning approaches. Generative AI can generate realistic content, figures, audio, and video, making it an impressive tool across multiple fields, including scientific research.⁶

In academic publishing, AI-assisted processes are progressively employed for data mining, analysis, document composition, and peer review, augmenting research efficiency and accessibility. As AI-driven technologies progress, concerns regarding reliability, ethical implications, and potential misuse in academic work have arisen.^{7,8} The emergence of AI-generated papers, data fabrication, and automated text manipulation have elicited apprehensions over the legitimacy of published research.^{9,10}

Retractions are essential for preserving the integrity of the scientific environment by officially eliminating published investigations that include errors, ethics incidents, or fraudulent data.¹¹ The retraction procedure generally commences after identifying substantial flaws in research. These flaws may arise from unforeseen mistakes or more serious concerns such as data manufacturing, plagiarism, and abuse of the peer review process.¹² Retractions serve the scientific community by averting the spread of erroneous or deceptive information. They maintain research integrity, preserve public confidence in science, and ensure that subsequent studies are not grounded on inaccurate data.¹³ The retraction process entails issuing a formal notification that elucidates the reasons for retraction and guarantees transparency. Notwithstanding retraction, citations to the retracted article can endure, underscoring the necessity for enhanced awareness and monitoring of retracted publications.¹⁴

While prior research has explored general retraction patterns, this study focuses solely on AI-related retractions, offering a new perspective on the ethical and methodological challenges facing this rapidly growing field. This study assesses AI-related retracted publications with the subsequent objectives:

- Examining the time-based distribution of retractions, encompassing peak years.
- Determining the countries experiencing the highest quantity of retracted AI-related papers.
- Classification of retraction rationale and identification of the predominant reasons.
- Evaluating retracted AI-related articles' citation and altmetric data to determine their influence.

This study seeks to identify the core reasons behind retractions, offering practical insights to strengthen editorial regulations, enhance the integrity of peer review, and ensure the responsible integration of AI into research.

METHODS

Study design

This descriptive cross-sectional investigation systematically discovered and assessed retracted publications related to AI without time limitations. A thorough evaluation was facilitated by conducting a search in PubMed with the query: “retracted publication”[publication type] [pt] and AI. The latest update of the acquired records was finalized on February 24, 2025. We utilized PubMed searches due to their straightforwardness and practicality, facilitating the swift retrieval of AI-related retracted publications. Moreover, PubMed was selected to maintain the analysis's focus on biomedical literature, thereby ensuring that the study remains pertinent to AI applications in healthcare and life sciences. This methodology enabled the incorporation of all relevant AI-related retracted publications accessible in the database up to the scheduled date. This thorough analysis provides valuable insights into the retraction of AI-related scholarly papers in the scientific literature.

Data extraction process

Bibliographic data from retracted AI-related publications were systematically obtained, exported, and classified in an Excel file for thorough review. This approach permitted adequate data storage and analysis, allowing a thorough examination of retraction dynamics. The data included fundamental publication characteristics:

- *Title*: The specific title of the retracted publication.
- *DOI (Digital Object Identifier)*: The distinctive alphanumeric sequence allocated to the paper facilitates accurate tracking and access.
- *Authors*: The complete list of researchers contributing to the publication.
- *Publication date*: The initial date of the article's release in the journal.
- *Retraction date*: The formal date when the journal or publisher officially retracts the publication.
- *Time to retractions*: The interval (in days) from initial publication to retraction, emphasizing the promptness of corrective measures.
- *Journal*: A scholarly source in which the academic work is published.
- *Article type*: The classification of the publication (e.g., original article, review, case report, editorial material, commentary).
- *Country of publication*: The country associated with the corresponding author, offering geographical context for retracted works.
- *Reason for retraction*: The rationale provided by the journal or publisher outlining the basis for the retraction.
- *Publisher*: The organization responsible for distributing the publication and managing its editorial norms.

Citation metrics were systematically obtained from the Web of Science database to evaluate the scholarly influence of the retracted publications. This allowed an assessment of the extent to which these retracted works were cited within the scientific community.

In addition, the journals that had initially published the retracted documents were categorized according to their indexing status in the Web of Science. In particular, they were categorized into the following indexing categories¹⁵:

- Science Citation Index Expanded (SCIE)
- Social Sciences Citation Index (SSCI)
- Emerging Sources Citation Index (ESCI)

Supplementary bibliometric metrics were documented to assess further the prominence and academic impact of the journals that published the retracted AI-related works. The quartile ranking and the 2023 impact factor of each journal were obtained from the Web of Science. Additionally, the journal categories were obtained from the Web of Science to ascertain the particular subject areas in which these journals are categorized.¹⁶

The Altmetric toolbar was utilized to acquire altmetric data. This tool offered insights into online interactions, reach, and public engagement across several platforms. The Altmetric Attention Score (AAS) is a crucial metric that measures the real-time attention a content has received. The AAS is a weighted composite score that quantifies the impact of an article by consolidating mentions from several internet sources. The AAS is computed with a proprietary technique that allocates distinct weights to various sources.^{17,18}

Categorization of retraction reasons

Retraction notices of retracted publications listed in PubMed were accessed to evaluate the rationales for the retraction. A systematic classification framework was utilized to categorize the reasons for retraction, ensuring a standardized and thorough review of the retracted publications. This classification was grounded in the arguments expressly articulated in the official retraction notices, allowing a clear comprehension of the various factors influencing the retraction process. The reasons for retraction were classified as follows¹⁹⁻²¹:

Error: This category included retractions caused by methodological defects such as issues with data collection, inadequate data representation, misinterpretation of outcomes, or improper research design. These defects could have resulted from honest mistakes or unintended oversights, jeopardizing the study's validity.

Fraud: Retractions in this category were due to deliberate scientific misconduct, including data falsification, fabrication, or manipulation. This featured purposeful changes to statistical information, graphics, figures, or experiment findings, all of which undermined study integrity and violated academic ethical standards.

Author disagreements: This category featured retractions caused by conflicts concerning authorship or intellectual contributions. Examples include scenarios when a study was published without the knowledge or approval of one or more participating researchers, including falsified authors or conflicts over project ownership between academics and funding agencies.

Duplication: This category pertains to cases when significantly similar or identical content was distributed in multiple journals, contravening publication ethics.

Ethics issues: Retractions within this group were linked to violations of ethical research norms, including the absence of informed consent from subjects, lack of approval from an institutional ethics review board, or overall non-compliance with established research ethics norms.

Peer review issues: Retractions resulting from peer review abnormalities, such as fraudulent, prejudiced, or manipulated review practices. This includes instances where reviewer identities were forged, conflicts of interest were not declared, or authors had undue influence over the review procedure.

Plagiarism: This category encompassed instances of unauthorized utilization of a different researcher's property rights, particularly text, charts, graphs, tables, methods of investigation, or hypotheses. Furthermore, self-plagiarism, defined as an author republishing their material without proper reference or disclosure, was similarly categorized.

Data concerns: Retractions were led by arguments regarding the validity, trustworthiness, or transparency of the data utilized throughout the research. Challenges with data accessibility, repeatability, and misleading statistical assessments also fell into this category.

Irrelevant citations: Publications that were retracted because of improper, exaggerated, or intentionally manipulative citations fell under this category. These manipulative tactics aimed to increase citation counts in a way that would benefit individual researchers or journals.

Unethical use of AI: This category includes retractions due to AI's incorrect or unethical use in research. Examples encompass unreported AI-generated content, the use of AI to falsify data or images, dependence on AI technologies for producing faked outcomes, and the omission of acknowledgment for AI-assisted writing or analysis where transparency is vital.

Unknown: Formal retraction notices failed to justify the retractions adequately. Labeling these cases as "unknown" acknowledges that the material lacked a clear justification.

To ensure a thorough and comprehensive examination of the factors contributing to retractions, each reason was documented and categorized separately when more than one retraction reason applied to a single publication.

Data processing

The data was rigorously formatted and analyzed in Microsoft Excel, ensuring accurate organization and effective handling. It was reported as numbers (n), percentages (%), and median values (minimum-maximum).

RESULTS

Study selection

A total of 778 documents were acquired from PubMed using the specified search settings. Publications without a definitive association with AI or inaccessible data were omitted from the study, and analyses were conducted on 764 articles.

Publication timeline and retraction trends

The initial retracted paper was published in 2002; the latest was published in 2024, spanning a publication range from 2002 to 2024. Retractions were first noted in 2003, and the most recent one occurred in 2025, representing the period from 2003 to 2025. **Fig. 1** illustrates that the annual frequency of retractions reached its highest point in 2023 (n = 667). The median duration from publication to retraction was 510 (28–4,200).

Country analysis

Fig. 2 shows the five countries with the highest number of retracted publications. China (n = 551) was the top country, followed by India (n = 40) in second place and Bangladesh (n = 23) in third. Saudi Arabia (n = 22) and Ethiopia (n = 20) followed these

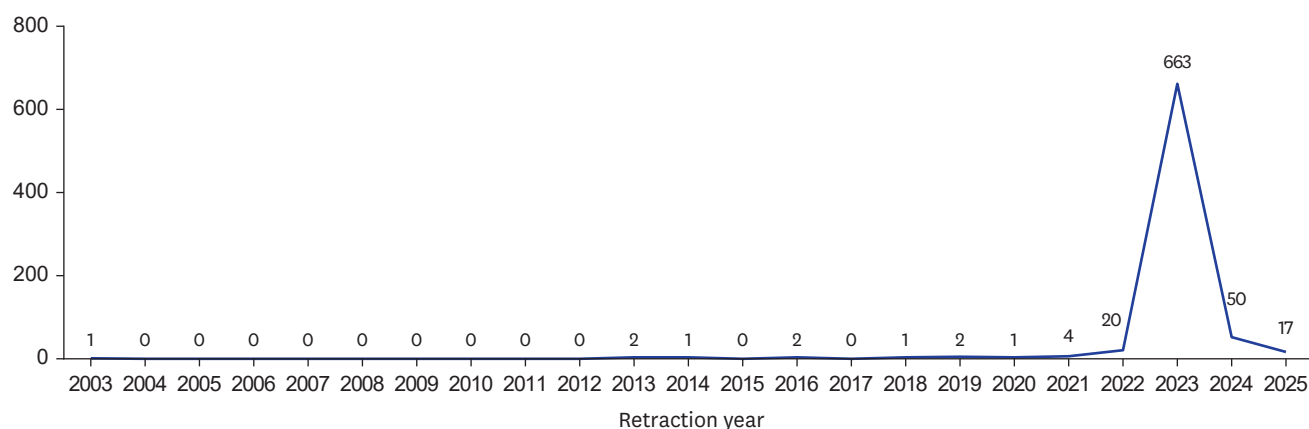


Fig. 1. Distribution of retracted publications by year.

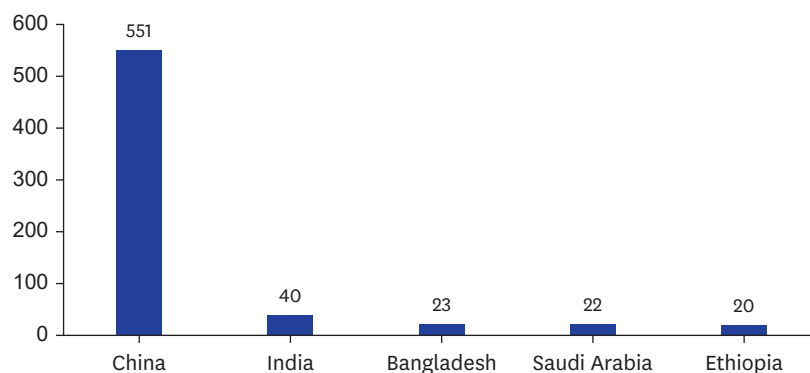


Fig. 2. Top five countries in terms of number of retracted publications.

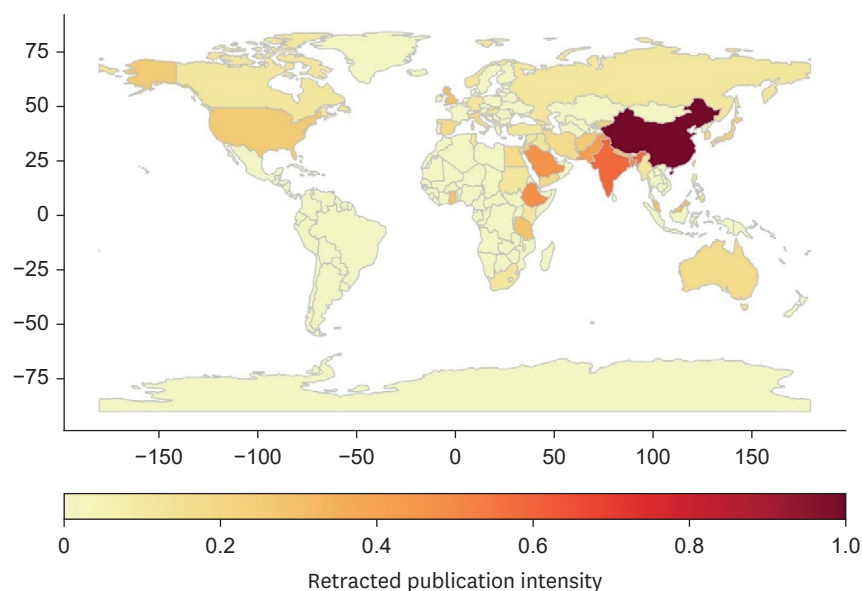


Fig. 3. Heatmap of the retracted publications.

countries. Fifteen publications had a corresponding author affiliated with Korea. Fig. 3 visualizes the country heat map.

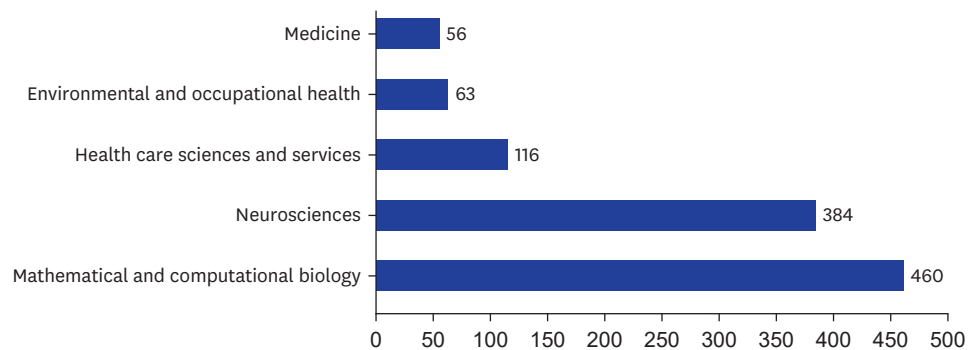


Fig. 4. Top five categories of journals that provide retracted publications.

Journal-based analysis

The ten journals with the highest number of retracted publications were the following: *COMPUTATIONAL INTELLIGENCE AND NEUROSCIENCE* (n = 383), *JOURNAL OF HEALTHCARE ENGINEERING* (n = 113), *COMPUTATIONAL AND MATHEMATICAL METHODS IN MEDICINE* (n = 75), *JOURNAL OF ENVIRONMENTAL AND PUBLIC HEALTH* (n = 62), *BIOMED RESEARCH INTERNATIONAL* (n = 47), *PLOS ONE* (n = 25), *NEUROSURGICAL REVIEW* (n = 9), *CONTRAST MEDIA & MOLECULAR IMAGING* (n = 8), *SCANNING* (n = 4), and *SCIENTIFIC REPORTS* (n = 4).

The categories of journals with retracted papers were documented using the Web of Science, and the analysis was performed by journal category. The top five categories were as follows: Mathematical and Computational Biology (n = 460), Neurosciences (n = 384), Health Care Sciences and Services (n = 116), Environmental and Occupational Health (n = 63), and Medicine (n = 56) (Fig. 4).

A total of 761 papers were published in SCIE-indexed journals, 101 were published in SSCI-indexed journals, and 3 were featured in the ESCI category.

Using Web of Science for quarterly analysis, 45 papers were published in Q1 journals, 588 in Q2 journals, 130 in Q3 journals, and one in a Q4 journal.

The median journal impact factor was 3.1 (0.8–29.9).

Publication types and citation data

Of the retracted publications, 735 were original articles, 18 were review papers, 8 were letters, and 3 were classified as editorial material. The median citation count for retracted papers was 2 (0–167).

Publisher data

The top three publishers in the analysis of retraction numbers of AI papers were HINDAWI (n = 673), PUBLIC LIBRARY OF SCIENCE (n = 25) and WILEY (n = 17).

Retraction reasons

Retraction reasons were categorized as outlined in the methodology. Each reason was recorded separately under the corresponding category when an article featured multiple retraction reasons. The following were the identified reasons for retraction: Peer review issues (n = 716), data concerns (n = 714), irrelevant citations (n = 571), unethical use of

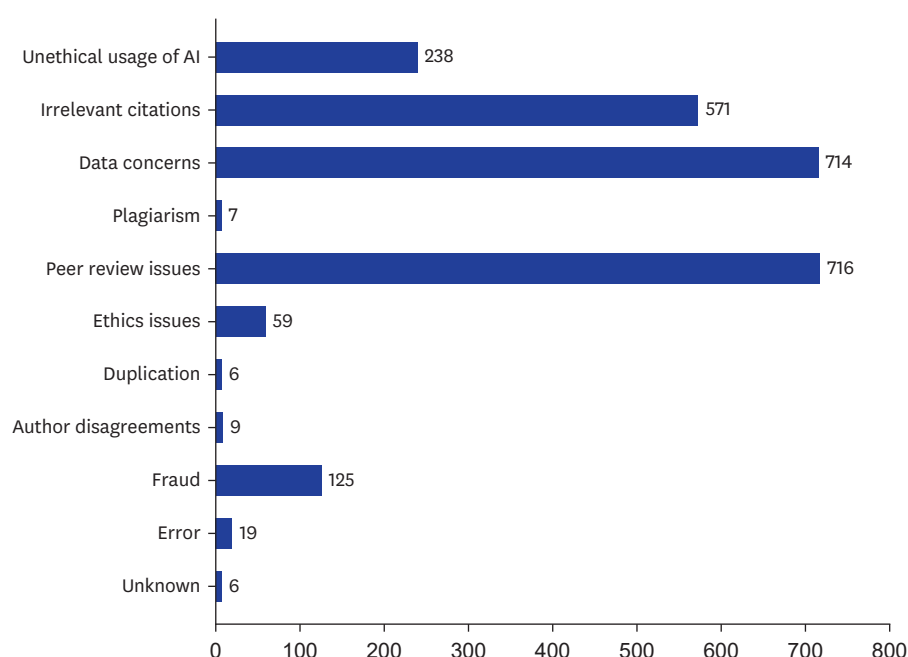


Fig. 5. Distribution of retraction reasons.
AI = artificial intelligence.

AI (n = 238), fraud (n = 125), ethics issues (n = 59), error (n = 19), author disagreements (n = 9), plagiarism (n = 7), duplication (n = 6), and unknown (n = 6) (Fig. 5).

Altmetric data

The median AAS value was 0 (0–191).

DISCUSSION

The investigation of AI-related retracted publications reveals crucial trends in scientific integrity and publishing ethics. The study indicated an increase in retractions, with 2023 (n = 663) having the highest number of retracted AI-related publications. China emerged as the foremost country with the most retracted publications, followed by India and Bangladesh. Journals devoted to mathematics, computational biology, neurosciences, and healthcare sciences had the highest retraction numbers. Peer review issues and data concerns were the most common reasons for retractions, followed by irrelevant citations and unethical AI use. The low median citation count and AAS reflect limited engagement with these papers, but the presence of highly cited retracted publications emphasizes the potential influence of misleading information.

The number of retracted publications has grown in recent years, and the high of retractions noted in 2023 indicates the strengthening of retraction standards, enhanced journal scrutiny, and a growing understanding of scientific misconduct. This tendency corresponds with wider advancements in academic publication, where apprehensions regarding research integrity have resulted in an increase in retractions across various disciplines.^{22,23} A further explanation for this outcome is the rising output of AI-related papers in recent years.²⁴ The number of retractions may rise concurrently with the growing volume of AI-related

publications. The median duration to retraction, 510 days, reveals a gap in detecting and fixing inaccurate papers, allowing potentially misleading findings to remain in the scientific environment. Given the rapid improvements in AI research, it is critical to shorten the lag period and implement more efficient post-publication review procedures to prevent the spread of erroneous or unethical content.²⁵

China's dominance in AI-related retractions, reaching 551 publications, far exceeds that of other countries, corroborating existing literature. Although India, Bangladesh, Saudi Arabia, and Ethiopia also contributed to the retracted literature, their roles were less numerous. The high volume of retracted AI-related papers from China, India, and other countries indicates pervasive systemic challenges in academic publishing and research integrity. China's prominence in retracted papers aligns with prior research emphasizing the rigorous "publish or perish" culture, wherein scholars face institutional pressures to obtain promotions, funding, and career advancement based on publishing statistics. This may result in an increase in questionable publication practices, including the use of paper mills, ghostwriting services, and manipulative peer reviews.²⁶⁻²⁸ In rapidly advancing research environments, quality control mechanisms struggle to keep pace with the surge in scientific production. The rise of predatory and substandard journals has exacerbated the issue, facilitating the inclusion of ethically dubious or low-quality studies.²⁹ These findings underscore the necessity for enhanced research integrity frameworks and ethical publishing standards, especially in countries with increased publication output.

The examination of retracted AI-related publications indicates that the majority were distributed in SCIE-indexed journals, signifying their inclusion in a reputable scientific setting. The discipline analysis indicates that retractions were most common in Mathematical and Computational Biology and Neurosciences, presumably because of the substantial use of AI-driven approaches in these fields. In particular, most retracted papers stemmed from Q2 journals, indicating that retraction issues are not limited to lower-tier publications.³⁰ The median journal impact factor of 3.1 suggests that retracted works were disseminated in journals of considerable prominence within their fields. Additionally, these findings highlight the necessity for stringent editorial supervision and enhanced peer review procedures throughout all journals.

The analysis of retraction reasons reveals that peer review issues ($n = 716$) and data concerns ($n = 714$) are the most common causes, underscoring ongoing challenges in upholding research integrity.^{31,32} Peer review is a voluntary process, and its effectiveness relies on the commitment of experts to invest time and effort in manuscript evaluation. The peer review process is the cornerstone of scientific publishing; therefore, enhancing its structure and making it more attractive to reviewers is essential.³³ The high number of retractions resulting from irrelevant citations suggests continued issues with citation manipulation, possibly motivated by unethical academic activities intended to enhance citation rankings artificially.³⁴ The presence of unethical AI use as a particular cause for retraction highlights increasing apprehensions about the ethical use of AI in scientific research. This category likely reflects instances of AI-generated products, dataset fabrication, or manipulated outcomes that are employed without proper disclosure or consideration of ethics.³⁵

Moreover, the results highlight a key issue that the scientific community should address: the integration of AI in research and its alignment with ethical principles. As AI technologies advance in sophistication, their application in research will progressively broaden.

Nonetheless, without strict ethical principles and transparency standards, the potential for misuse persists. The scientific community should prioritize establishing explicit regulations and optimal procedures to guarantee the ethical utilization of AI, thereby preserving the integrity and legitimacy of academic research.³⁶

The relatively low median citation and AAS values suggest that most of these papers had limited scholarly and public impact. While specific, highly referenced, or frequently discussed, retracted works show that incorrect findings can obtain momentum. Retracted yet extensively cited publications are a significant concern, especially in the field of AI, where erroneous or unethical outcomes can impact future research, clinical decision-making, and policy-making. The prolonged existence of retracted publications in citation networks can lead to the dissemination of misleading findings, thereby jeopardizing scientific integrity and undermining evidence-based procedures. These findings underscore the importance of effective retraction methods in preventing the spread of misinformation in both the scientific and public spheres.

Journal editors must enforce rigorous protocols during the peer review and in-house assessment phases to mitigate the risk of retractions. A crucial strategy is to mandate disclaimers concerning the ethical utilization of AI and AI-assisted editing tools at the time of submission. Authors must disclose the use of AI in data analysis, manuscript composition, or processing of images to ensure responsibility and traceability.³⁷

A critical outcome of this study was that peer review issues were the most common reason for retraction, suggesting systemic flaws in the article assessment process. Many retracted publications were most likely subjected to substandard or compromised peer review, which fake peer reviewers could have caused, suggested reviewers with conflicts of interest, or paper mills abusing editorial vulnerabilities. Furthermore, specific authors may have intentionally manipulated the peer review process to ensure their submissions were easily accepted and published. This could include suggesting favorable or fabricated reviewers, presenting to journals with known poor editorial control or exploiting peer review gaps to speed acceptance. To tackle these issues, journals should implement more stringent reviewer selection methods, assure independent validation of suggested reviewers, and deliver peer review education. Furthermore, editors should thoroughly examine raw data and methods details, ensuring the legitimacy and reproducibility of presented results.³⁸

This study has several limitations that require consideration. The investigation primarily relied on data from PubMed, potentially omitting retracted papers predominantly indexed in alternative databases. A notable limitation is language restriction, as the study mainly utilized PubMed, which predominantly indexes documents in English. This may have resulted in an insufficient representation of retractions from non-English sources. The variability in the formatting of retraction notices across different journals may have impacted the consistency of data collection. The results signify a particular moment in time, and it is crucial to note that the data may fluctuate as additional retractions transpire and the academic publishing scene evolves.

This study provides valuable insights into the retractions of AI-related papers. The growing number of retracted AI publications in recent years, particularly in 2023, indicates an escalating awareness of research integrity challenges. The large number of retractions in China highlights the need for enhanced research standards, particularly in areas

with a substantial publication volume. Peer review gaps and data concerns surfaced as the predominant causes for retraction, highlighting ongoing challenges in upholding research integrity and quality assurance. The substantial number of retractions, particularly from certain countries and publishers, raises concerns about the potential impact of paper mills, which generate and sell fraudulent academic papers. Although our research did not specifically analyze paper mills, the patterns identified, such as large retractions from particular journals and publishers, are consistent with previously documented instances of paper mill activity.³⁹ The emergence of unethical AI use underscores the necessity to establish comprehensive criteria for the ethical application of AI in research. Enhanced author guidelines should establish explicit protocols for the ethical utilization of AI, guaranteeing transparency in the role of AI tools in research, analysis, and content development. Both in-house and external peer review processes should integrate advanced AI-detection systems to ensure the integrity of AI-generated material, therefore mitigating risks of bias, fabrication, or misinformation. All submissions require mandatory disclaimers that indicate whether AI was utilized in the paper's design, authoring, or editing. These strategies will promote responsible AI utilization, maintain research integrity, and reduce the likelihood of retractions.^{40,41}

REFERENCES

1. Rashid AB, Kausik AK. AI revolutionizing industries worldwide: a comprehensive overview of its diverse applications. *Hybrid Advances* 2024;7:100277. [CROSSREF](#)
2. Alowais SA, Alghamdi SS, Alsuhbany N, Alqahtani T, Alshaya AI, Almohareb SN, et al. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Med Educ* 2023;23(1):689. [PUBMED](#) | [CROSSREF](#)
3. Jiang F, Jiang Y, Zhi H, Dong Y, Li H, Ma S, et al. Artificial intelligence in healthcare: past, present and future. *Stroke Vasc Neurol* 2017;2(4):230-43. [PUBMED](#) | [CROSSREF](#)
4. Guo Y, Hao Z, Zhao S, Gong J, Yang F. Artificial intelligence in health care: bibliometric analysis. *J Med Internet Res* 2020;22(7):e18228. [PUBMED](#) | [CROSSREF](#)
5. Medical Subject Headings (MeSH). Artificial intelligence. <https://www.ncbi.nlm.nih.gov/mesh/68001185>. Updated 1986. Accessed March 10, 2025.
6. Medical Subject Headings (MeSH). Generative artificial intelligence. <https://www.ncbi.nlm.nih.gov/mesh/2108164>. Updated 2025. Accessed March 10, 2025.
7. Doskaliuk B, Zimba O, Yessirkepov M, Klishch I, Yatsyshyn R. Artificial intelligence in peer review: enhancing efficiency while preserving integrity. *J Korean Med Sci* 2025;40(7):e92. [PUBMED](#) | [CROSSREF](#)
8. Kocak Z. Publication ethics in the era of artificial intelligence. *J Korean Med Sci* 2024;39(33):e249. [PUBMED](#) | [CROSSREF](#)
9. Guleria A, Krishan K, Sharma V, Kanchan T. ChatGPT: ethical concerns and challenges in academics and research. *J Infect Dev Ctries* 2023;17(9):1292-9. [PUBMED](#) | [CROSSREF](#)
10. Kocyigit BF, Zhaksylyk A. Advantages and drawbacks of ChatGPT in the context of drafting scholarly articles. *Cent Asian J Med Hypotheses Ethics* 2023;4(3):163-7. [CROSSREF](#)
11. Kocyigit BF, Akyol A. Analysis of retracted publications in the biomedical literature from Turkey. *J Korean Med Sci* 2022;37(18):e142. [PUBMED](#) | [CROSSREF](#)
12. Kocyigit BF, Zhaksylyk A, Akyol A, Yessirkepov M. Characteristics of retracted publications from Kazakhstan: an analysis using the retraction watch database. *J Korean Med Sci* 2023;38(46):e390. [PUBMED](#) | [CROSSREF](#)
13. Moylan EC, Kowalczyk MK. Why articles are retracted: a retrospective cross-sectional study of retraction notices at BioMed Central. *BMJ Open* 2016;6(11):e012047. [PUBMED](#) | [CROSSREF](#)
14. Khan H, Gasparyan AY, Gupta L. Lessons learned from publicizing and retracting an erroneous hypothesis on the mumps, measles, rubella (MMR) vaccination with unethical implications. *J Korean Med Sci* 2021;36(19):e126. [PUBMED](#) | [CROSSREF](#)
15. Prancutė R. Web of Science (WoS) and Scopus: the titans of bibliographic information in today's academic world. *Publications* 2021;9(1):12. [CROSSREF](#)

16. Atallah AN, Puga MEDS, Amaral JLGD. Web of Science journal citation report 2020: the Brazilian contribution to the "Medicine, General & Internal" category of the journal impact factor (JIF) ranking (SCI 2019). *Sao Paulo Med J* 2020;138(4):271-4. [PUBMED](#) | [CROSSREF](#)
17. Kocyigit BF, Akyol A. Bibliometric and altmetric analyses of publication activity in the field of Behcet's disease in 2010-2019. *J Korean Med Sci* 2021;36(32):e207. [PUBMED](#) | [CROSSREF](#)
18. Koçyiğit BF, Akyol A. Altmetrics and citation metrics as complementary indicators for research management. *Cent Asian J Med Hypotheses Ethics* 2021;2(2):79-84. [CROSSREF](#)
19. Stavale R, Ferreira GI, Galvão JAM, Zicker F, Novaes MRCG, Oliveira CM, et al. Research misconduct in health and life sciences research: a systematic review of retracted literature from Brazilian institutions. *PLoS One* 2019;14(4):e0214272. [PUBMED](#) | [CROSSREF](#)
20. Resnik DB, Hosseini M. The ethics of using artificial intelligence in scientific research: new guidance needed for a new tool. *AI Ethics* 2025;5(2):1499-521. [PUBMED](#) | [CROSSREF](#)
21. Campos-Varela I, Ruano-Raviña A. Misconduct as the main cause for retraction. A descriptive study of retracted publications and their authors. *Gac Sanit* 2019;33(4):356-60. [PUBMED](#) | [CROSSREF](#)
22. Song F, Wu B, Wei G, Cheng S, Wei L, Xiong W, et al. A systematic analysis of temporal trends, characteristics, and citations of retracted stem cell publications. *BMC Med* 2025;23(1):131. [PUBMED](#) | [CROSSREF](#)
23. Qi Q, Huang J, Wu Y, Pan Y, Zhuang J, Yang X. Recent trends: retractions of articles in the oncology field. *Heliyon (Lond)* 2024;10(12):e33007. [PUBMED](#) | [CROSSREF](#)
24. Lu W, Yu X, Li Y, Cao Y, Chen Y, Hua F. Artificial intelligence-related dental research: bibliometric and altmetric analysis. *Int Dent J* 2025;75(1):166-75. [PUBMED](#) | [CROSSREF](#)
25. Decullier E, Huot L, Maisonneuve H. What time-lag for a retraction search on PubMed? *BMC Res Notes* 2014;7(1):395. [PUBMED](#) | [CROSSREF](#)
26. Khademizadeh S, Danesh F, Esmaeili S, Lund B, Santos-d'Amorim K. Evolution of retracted publications in the medical sciences: citations analysis, bibliometrics, and altmetrics trends. *Account Res* 2024;31(8):1182-97. [PUBMED](#) | [CROSSREF](#)
27. Shi L, Zhang X, Ma X, Sun X, Li J, He S. Mapping retracted articles and exploring regional differences in China, 2012-2023. *PLoS One* 2024;19(12):e0314622. [PUBMED](#) | [CROSSREF](#)
28. Zhaksylyk A, Zimba O, Yessirkepov M, Kocyigit BF. Research integrity: where we are and where we are heading. *J Korean Med Sci* 2023;38(47):e405. [PUBMED](#) | [CROSSREF](#)
29. Althaus F, Kohl CBS, Faggion CM Jr. An overview of studies assessing predatory journals within the biomedical sciences. *Account Res* 2025;24:1-20. [PUBMED](#) | [CROSSREF](#)
30. Khurana P, Sharma K, Uddin Z. Unraveling retraction dynamics in COVID-19 research: patterns, reasons, and implications. *Account Res* 2024;23:1-24. [PUBMED](#) | [CROSSREF](#)
31. Ghorbi A, Fazeli-Varzaneh M, Ghaderi-Azad E, Ausloos M, Kozak M. Retracted papers by Iranian authors: causes, journals, time lags, affiliations, collaborations. *Scientometrics* 2021;126(9):7351-71. [CROSSREF](#)
32. Gedik MS, Kaya E, Kilci AI. Evaluation of retracted articles in the field of emergency medicine on the web of science database. *Am J Emerg Med* 2024;82:68-74. [PUBMED](#) | [CROSSREF](#)
33. Gasparyan AY, Gerasimov AN, Voronov AA, Kitas GD. Rewarding peer reviewers: maintaining the integrity of science communication. *J Korean Med Sci* 2015;30(4):360-4. [PUBMED](#) | [CROSSREF](#)
34. Gasparyan AY, Yessirkepov M, Voronov AA, Gerasimov AN, Kostyukova EI, Kitas GD. Preserving the integrity of citations and references by all stakeholders of science communication. *J Korean Med Sci* 2015;30(11):1545-52. [PUBMED](#) | [CROSSREF](#)
35. Chetwynd E. Ethical use of artificial intelligence for scientific writing: current trends. *J Hum Lact* 2024;40(2):211-5. [PUBMED](#) | [CROSSREF](#)
36. Samuel G, Chubb J, Derrick G. Boundaries between research ethics and ethical research use in artificial intelligence health research. *J Empir Res Hum Res Ethics* 2021;16(3):325-37. [PUBMED](#) | [CROSSREF](#)
37. Hosseini M, Resnik DB, Holmes K. The ethics of disclosing the use of artificial intelligence tools in writing scholarly manuscripts. *Res Ethics Rev* 2023;19(4):449-65. [PUBMED](#) | [CROSSREF](#)
38. Barroga E. Innovative strategies for peer review. *J Korean Med Sci* 2020;35(20):e138. [PUBMED](#) | [CROSSREF](#)
39. Rivera H, Teixeira da Silva JA. Retractions, fake peer reviews, and paper mills. *J Korean Med Sci* 2021;36(24):e165. [PUBMED](#) | [CROSSREF](#)
40. AlSamhori ARF, Alnaimat F. Artificial intelligence in writing and research: ethical implications and best practices. *Cent Asian J Med Hypotheses Ethics* 2024;5(4):259-68. [CROSSREF](#)
41. Habibzadeh F. GPTZero performance in identifying artificial intelligence-generated medical texts: a preliminary study. *J Korean Med Sci* 2023;38(38):e319. [PUBMED](#) | [CROSSREF](#)